

# A $\sigma$ -Essential Contextual State: Localizing an Open Problem on Concrete Orthomodular Lattices

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## Abstract

A two-valued state on a concrete orthomodular lattice (OML) of subsets of a set  $\Omega$  assigns a definite truth value to each measurable event, coherently across overlapping measurement contexts. On a Boolean algebra every finitely coherent local pattern of such assignments extends to a global  $\sigma$ -additive two-valued state, and (away from a measurable cardinal) every global one is a point evaluation. We ask whether the non-distributive case can differ: does there exist a concrete  $\sigma$ -complete non-Boolean OML carrying a finitely coherent local pattern that extends to *no* global  $\sigma$ -additive two-valued state — a  *$\sigma$ -essential contextual state*? We show this question *localizes*: a witness exists iff (i) no point evaluation extends the pattern and (ii) no non-principal  $\sigma$ -additive two-valued state does either, and that clause (i) is freely arrangeable (after Navara–Pták), so the entire content sits in clause (ii). We then strip clause (ii) of all lattice scaffolding to a bare set-theoretic existence statement — a non-principal,  $\sigma$ -additive, coherent selection across a system of overlapping countable partitions — and locate it precisely against the surrounding literature: it is empty in the Polish-representable case (Derr–Williamson), it does *not* reduce to the Hilbert-space pure-state dichotomy of Blecher–Weaver (Kochen–Specker; Akemann–Weaver), and it does not reduce to a measurable cardinal in the way its Boolean shadow would. The result is a sharply posed, currently open problem at the interface of quantum logic and the set theory of measure-theoretic structures, together with a map of where its answer must live.

## 1 Introduction

A finite system of measurement contexts may admit a consistent assignment of truth values to events that no single underlying state realises — the algebraic signature of quantum contextuality. When the contexts are countable and one demands countable additivity, a sharper question arises: can a *finitely* coherent pattern fail to extend to any global  $\sigma$ -additive state, with the failure visible only at the  $\sigma$ -level? This paper isolates that phenomenon for concrete orthomodular lattices, reduces it to an irreducible set-theoretic core, and locates that core against the measure-theoretic and operator-algebraic literature. We work throughout with two-valued states, the non-distributive analogue of a  $\{0, 1\}$ -valued measure.

A *concrete  $\sigma$ -complete orthomodular lattice* (concrete  $\sigma$ -OML) is a family  $\mathcal{L} \subseteq \mathcal{P}(\Omega)$  with

$$\emptyset, \Omega \in \mathcal{L}; \quad A \in \mathcal{L} \implies A^\perp := \Omega \setminus A \in \mathcal{L}; \quad \{A_n\} \subseteq \mathcal{L} \text{ pairwise disjoint} \implies \bigcup_n A_n \in \mathcal{L},$$

ordered by inclusion. Equivalently it is a  *$\sigma$ -class* [17], a Gudder concrete quantum logic [10, 18]: the maximal Boolean subalgebras (“blocks”) overlap but their union need not be Boolean. A *two-valued state* is  $s: \mathcal{L} \rightarrow \{0, 1\}$  with  $s(\Omega) = 1$  that is additive over orthogonal joins lying in its domain: whenever  $\{A_n\} \subseteq \mathcal{L}$  are pairwise disjoint with  $\bigcup_n A_n \in \mathcal{L}$ ,

$$s\left(\bigcup_n A_n\right) = \sum_n s(A_n)$$

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(in particular  $s(A) + s(A^\perp) = 1$  on complement pairs); it is  $\sigma$ -additive when this holds for countable such families. The same additivity is required of states on a sub-orthoposet  $B \subseteq \mathcal{L}$ , over orthogonal joins that exist in  $B$ . The *point evaluations*  $\delta_\omega(A) := \mathbb{1}[\omega \in A]$  are  $\sigma$ -additive two-valued states; call these *Dirac* and all others *non-Dirac*. For a finite  $\perp$ -closed sub-orthoposet  $B \subseteq \mathcal{L}$  and a two-valued state  $s_0$  on  $B$ ,  $s_0$  *extends* to  $s$  on  $\mathcal{L}$  if  $s \upharpoonright B = s_0$ .

**Definition 1.1** ( $\sigma$ -essential contextual state). A  $\sigma$ -essential contextual state on  $\mathcal{L}$  is a two-valued state  $s_0$  on a finite  $\perp$ -closed sub-orthoposet  $B \subseteq \mathcal{L}$  that extends to *no* global  $\sigma$ -additive two-valued state on  $\mathcal{L}$ . A concrete  $\sigma$ -OML  $\mathcal{L}$  *carries a  $\sigma$ -essential contextual state* (is a *witness*) if such an  $s_0$  exists. We require  $\mathcal{L}$  to be non-Boolean and *irreducible* (centre  $\{\emptyset, \Omega\}$ ); see Remark 1.2.

The phenomenon is a failure of  $\sigma$ -level compactness. Wright’s theorem [20] — every two-valued state on a *finite* OML is finitely realised — forbids any *finite* witness, so the question is whether a coherent finite pattern can fail to globalise once  $\sigma$ -additivity is imposed. On a Boolean algebra it cannot (§2); we isolate what must differ in the non-distributive case, reduce to the irreducible core, and locate it.

**Results.** §2: the Boolean baseline (Prop. 2.1, two obstructions). §3: the *Localization Theorem* (Thm. 3.2) — a witness exists iff (i) no Dirac and (ii) no non-Dirac  $\sigma$ -additive two-valued state extends  $s_0$ , with (i) freely arrangeable, localising all weight to (ii), which strips to a bare set-theoretic selection problem (Q. 3.6). §4: the boundary map — empty above the Polish case, irreducible to Blecher–Weaver, irreducible to a measurable cardinal. §5: the open problem, its target sentence, and the constraints localising it.

**Remark 1.2** (Irreducibility is not vacuous). The carrier class is inhabited: the horizontal sum  $MO_\kappa$  ( $\kappa$  orthogonal atom-pairs with 0, 1) is concrete, non-Boolean, and irreducible (centre  $\{\emptyset, \Omega\}$ ) for  $\kappa \geq 2$  [13], and  $\sigma$ -complete for  $\kappa = \omega$ . Concreteness does not force a non-trivial centre: by the Zierler–Schlessinger/Gudder criterion it is exactly order-determination by two-valued states [10, 11], unrelated to reducibility. ( $MO_\kappa$  itself is excluded as a witness by Remark 4.2.)

## 2 The Boolean baseline

On a Boolean  $\sigma$ -algebra, Definition 1.1 is never satisfied. The proof isolates two distinct obstructions a witness must overcome at once; this is the baseline the non-distributive case must escape.

**Proposition 2.1** (No witness in the Boolean case). *Let  $\mathcal{L} \subseteq \mathcal{P}(\Omega)$  be a Boolean  $\sigma$ -algebra and  $B \subseteq \mathcal{L}$  a finite  $\perp$ -closed sub-orthoposet with a two-valued state  $s_0$ . Then  $s_0$  extends to a global  $\sigma$ -additive two-valued state on  $\mathcal{L}$ .*

*Proof.* Let  $B_0 \subseteq \mathcal{L}$  be the finite Boolean subalgebra generated by  $B$ . Because  $\mathcal{L}$  is Boolean and  $s_0$  is additive over the orthogonal joins in  $B$ ,  $s_0$  extends to a (finitely additive) two-valued state on  $B_0$ : on a finite Boolean algebra such a state is determined by, and concentrates on, a single atom  $a \in B_0$  — the meet of the  $s_0$ -true generators. As  $a$  is a nonzero element of the concrete algebra  $B_0$ , it is a nonempty subset of  $\Omega$ , and

$$K(s_0) := \bigcap \{A \in B : s_0(A) = 1\} \supseteq a \neq \emptyset$$

(we call  $K(s_0)$  the *kernel* of  $s_0$ ). Any  $\omega \in K(s_0)$  satisfies  $\delta_\omega \upharpoonright B = s_0$  by Lemma 3.1, and  $\delta_\omega$  is  $\sigma$ -additive. So  $s_0$  globalises — and on a concrete Boolean  $\sigma$ -algebra it does so by a Dirac, the case  $K(s_0) = \emptyset$  never arising. The extension to  $B_0$  is exactly where distributivity is used; it is unavailable on a non-Boolean carrier, where the  $s_0$ -true generators may have empty meet.  $\square$

Two obstructions combine, both of which a witness must defeat:

- (a) (*used above*) compatible events of  $B$  generate a finite Boolean algebra whose  $s_0$ -carrying atom is a nonempty common point. A witness breaks this — the events lie in incompatible blocks with empty meet, so no Dirac extends  $s_0$ .
- (b) (*complementary*) away from a measurable cardinal, every global  $\sigma$ -additive two-valued state on a Boolean  $\sigma$ -algebra is Dirac [19, 12]; so even where (a) fails, no *non*-Dirac state repairs the gap on a Boolean carrier.

The two pull against each other: breaking (a) requires non-distributivity, which is precisely what makes (b) escapable by admitting non-Dirac global states. A witness must break (a) while leaving no such repair available.

### 3 The Localization Theorem

When a Dirac extends  $s_0$  is elementary.

**Lemma 3.1.** *Let  $B \subseteq \mathcal{L}$  be a finite  $\perp$ -closed sub-orthoposet with two-valued state  $s_0$ . A point evaluation  $\delta_\omega$  satisfies  $\delta_\omega \upharpoonright B = s_0$  iff  $\omega \in K(s_0) = \bigcap \{A \in B : s_0(A) = 1\}$ . Consequently no Dirac extends  $s_0$  iff  $K(s_0) = \emptyset$ .*

*Proof.*  $\delta_\omega \upharpoonright B = s_0$  means  $\omega \in A \iff s_0(A) = 1$  for every  $A \in B$ . The forward constraints place  $\omega \in \bigcap \{A : s_0(A) = 1\}$ . For a false  $A$  (i.e.  $s_0(A) = 0$ ),  $\perp$ -closure gives  $A^\perp \in B$ , and additivity gives  $s_0(A^\perp) = 1 - s_0(A) = 1$ ; so  $A^\perp$  is one of the true elements and the constraint  $\omega \notin A$  is already  $\omega \in A^\perp$ , subsumed in  $K(s_0)$ . Thus the false constraints add nothing, and  $\delta_\omega \upharpoonright B = s_0 \iff \omega \in K(s_0)$ . Conversely any  $\omega \in K(s_0)$  works: true  $A$  give  $\omega \in A$ , false  $A$  give  $\omega \in A^\perp$  hence  $\omega \notin A$ .  $\square$

**Theorem 3.2** (Localization). *Let  $\mathcal{L}$  be a concrete  $\sigma$ -OML,  $B \subseteq \mathcal{L}$  a finite  $\perp$ -closed sub-orthoposet, and  $s_0$  a two-valued state on  $B$ . Then  $s_0$  is a  $\sigma$ -essential contextual state iff*

- (i)  $K(s_0) = \emptyset$  (no point evaluation extends  $s_0$ ), and
- (ii) no non-Dirac  $\sigma$ -additive two-valued state on  $\mathcal{L}$  extends  $s_0$ .

Moreover clause (i) is freely arrangeable: there is a concrete  $\sigma$ -OML and an  $s_0$  with  $K(s_0) = \emptyset$ .

*Proof.* Every global  $\sigma$ -additive two-valued state on  $\mathcal{L}$  is either a Dirac  $\delta_\omega$  or non-Dirac, and this dichotomy is exhaustive. Hence “no global  $\sigma$ -additive two-valued state extends  $s_0$ ” is equivalent to the conjunction “no Dirac extends  $s_0$ ” and “no non-Dirac extends  $s_0$ .” By Lemma 3.1 the first conjunct is (i); the second is (ii). For arrangeability of (i), the Navara–Pták construction (Example 3.4) realises  $K(s_0) = \emptyset$  explicitly.  $\square$

**Remark 3.3** (Scope: a decomposition, not a reduction). The equivalence is a decomposition, not a reduction to a separate named problem: the Dirac/non-Dirac split being exhaustive, it merely unfolds the definition of witness. Its content is that clause (i) is *freely arrangeable*, so all weight of Definition 1.1 sits in clause (ii) — the  $\sigma$ -point-selection core, the subject of the remainder.

The Navara–Pták construction both supplies clause (i) and shows (ii) is binding.

**Example 3.4** (Navara–Pták [17]). Let  $\Omega = \mathbb{Q}_0 \times \mathbb{Q}_0$  with  $\mathbb{Q}_0 = \mathbb{Q} \cap (0, 1)$ , and let  $f, g: \Omega \rightarrow \mathbb{R}$  be the coordinate projections. Let  $\mathcal{L} = \mathcal{L}_{f,g}$  be the least  $\sigma$ -class containing the inverse images of Borel sets under  $f$  and  $g$ . Set  $C_f = f^{-1}(\{1/3\})$ ,  $C_g = g^{-1}(\{1/2\})$ ,  $C_{f+g} = (f+g)^{-1}(\{1/2\})$ . Define  $m(A) = 1$  iff  $A$  contains one of  $C_f, C_g, C_{f+g}$ . Navara and Pták show  $m$  is a (two-valued,  $\sigma$ -additive) state on  $\mathcal{L}$ , and that since  $C_f \cap C_g \cap C_{f+g} = \emptyset$  the state  $m$  is *not* concentrated at any point.

**Remark 3.5** (The example is not itself a witness). Let  $B$  be the  $\perp$ -closure of  $\{C_f, C_g, C_{f+g}\}$  and let  $s_0$  be the pattern assigning each of the three the value 1. Since  $C_f \cap C_g \cap C_{f+g} = \emptyset$ , we have  $K(s_0) = \emptyset$ , so clause (i) holds — no point evaluation extends  $s_0$ . Yet the Navara–Pták state satisfies  $m \upharpoonright B = s_0$  and is non-Dirac and  $\sigma$ -additive, so clause (ii) *fails* and  $s_0$  is not  $\sigma$ -essential: the construction supplies its own rescuer, so clause (i) cannot suffice alone. The repair is available because  $\mathcal{L}_{f,g}$  is generated by two *compatible* (commuting) observables — Boolean enough to support a non-Dirac global state through the point-level gap. A witness requires the opposite: a carrier on which no non-Dirac state repairs the gap.

Stripped of lattice scaffolding, this core — clause (ii) — is a problem about selections. A two-valued state selects one “true” cell within each block, a maximal compatible context, equivalently a countable partition of  $\Omega$  into elements of  $\mathcal{L}$ . Three conditions encode the structure: overlapping blocks agree on shared events (coherence);  $\sigma$ -additivity constrains countable joins; and non-Diracness means no single point realises the selection.

**Question 3.6** ( $\sigma$ -point-selection, bare form). Given a set  $\Omega$  and a family  $\{\Pi_\alpha\}$  of countable partitions of  $\Omega$  (the blocks of a concrete  $\sigma$ -OML, pairwise overlapping in their common refinements), does there exist a selection assigning to each  $\Pi_\alpha$  one cell  $c_\alpha \in \Pi_\alpha$  such that

- (a) (coherence) the selections agree on events common to several blocks;
- (b) ( $\sigma$ -additivity) the induced  $\{0, 1\}$ -valuation commutes with countable disjoint unions;
- (c) (non-principality) there is no  $\omega_0 \in \Omega$  with  $c_\alpha = \omega_0$  for all  $\alpha$ ;
- (d) (prescription) the selection realises a prescribed finite pattern that no single point realises?

A witness exists iff Question 3.6 has a positive answer for some admissible carrier. Two features of the bare form, obscured in lattice language, fix its position: its distance from a measurable cardinal, and its place within contextuality.

**Remark 3.7** (The Boolean shadow is a measurable cardinal; the question is not). For a *compatible* block family (single Boolean base), a coherent non-principal  $\sigma$ -additive  $\{0, 1\}$ -selection = a non-principal countably-complete ultrafilter = a measurable cardinal [12]. Question 3.6 imposes coherence (a) across *incompatible, mutually non-distributive* blocks — strictly stronger than an ultrafilter on any single block. So the *Boolean shadow* reduces to a measurable cardinal, but the non-distributive coherence is exactly what the reduction cannot cross (§4).

**Remark 3.8** (Relation to sheaf-theoretic contextuality). Question 3.6 asserts that locally coherent  $\{0, 1\}$ -valuations admit no *global section* — contextuality in the sense of Abramsky–Brandenburger [2],  $\sigma$ -additive form [1]. The framing is standard; what is open is the conjunction the literature does not treat together: countable additivity + a non-Hilbert (abstractly concrete) carrier + a  $\{0, 1\}$ -valued selection.

## 4 The boundary map

We locate Question 3.6 between an upper boundary where it is known empty and the lower routes that fail to settle it.

The upper boundary is the Polish-representable case. Derr and Williamson [6] prove a  $\sigma$ -extension criterion for coherent probabilities on pre-Dynkin systems: when the carrier is *Polish-representable* —  $\Omega$  Polish, the generated  $\sigma$ -class equal to the Borel  $\sigma$ -algebra, with inner-regular blocks — finite coherence implies  $\sigma$ -extendability (their Theorem D.6, via Maharam [16]). Specialised to two-valued states, this says that on a Polish-representable carrier every finitely coherent  $s_0$  globalises: *no  $\sigma$ -essential contextual state exists*. A witness, if one exists, must therefore live on a carrier that is *not* Polish-representable.

A Hilbert-space analogue exists, but does not transfer down to the concrete case. On the projection lattice of  $B(\ell^2(\kappa))$ , Blecher–Weaver [5] show a singular countably additive *pure* state exists iff  $\kappa$  is Ulam measurable — the nearest positive precedent that countable additivity on a non-distributive  $\sigma$ -complete OML *can* carry set-theoretic strength. It does not settle Question 3.6; three obstructions point the same way.

- (1) *Wrong carrier.* For  $\dim H \geq 3$  the projection lattice of  $B(H)$  admits *no* two-valued state (Kochen–Specker [14]), so it is not concrete; the Blecher–Weaver dichotomy concerns *pure* states, not two-valued ones, which are absent there by Gleason [9].
- (2) *Masa transfer fails.* Carrying a state-existence result to an abelian subalgebra means factoring through a masa; for the pure two-valued case this fails — Akemann–Weaver [3] give a pure state on  $B(H)$  multiplicative on no masa, and Blecher–Weaver use Marcus–Spielman–Srivastava paving for exactly this reason.
- (3) *The consumed object is structurally absent.* Blecher–Weaver extract their cardinal from a genuine ultrafilter on the diagonal masa  $\ell^\infty(\kappa)$  (projections = subsets, intersection-closed). On a concrete non-Boolean  $\sigma$ -OML,  $\mathcal{F}_s := \{A \in \mathcal{L} : s(A) = 1\}$  is an ultrafilter for *no* two-valued state  $s$  — not even a filter: meet-closure fails already for  $\delta_\omega$ , since in any concrete  $MO_2$  the atoms  $a, b$  have  $a \cap b \neq \emptyset$  (else orthogonal) yet  $a \wedge b = 0$ , so

$$\delta_\omega(a) = \delta_\omega(b) = 1, \quad \delta_\omega(a \wedge b) = 0.$$

The countably-complete ultrafilter an Ulam-style argument would consume is absent from the carrier — the same non-distributivity (lattice meet  $\neq$  set intersection), one level down.

**Remark 4.1** (Strength of the claim). Remark 3.7 and (1)–(3) are *cited facts assembled*, not a non-reducibility theorem in a fixed reduction calculus. The claim is only that the two obvious bridges — “measurable cardinal” and “transfer from Blecher–Weaver via a masa” — do not close Question 3.6, the obstruction in each being the same non-distributivity. This is what leaves the question open rather than a corollary.

The remaining route from below, faithful representation, is barred by the same obstruction. Routing Question 3.6 through a faithful  $\sigma$ -tribe representation (embedding  $\mathcal{L}$  into a  $\sigma$ -algebra of sets, so two-valued states become point evaluations) is unavailable: a faithful tribe-of-sets representation forces distributivity, hence Booleanness (the “Floor” [18]).  $\sigma$ -Loomis–Sikorski holds under the Riesz Decomposition Property ( $\sigma$ -complete MV-algebras, RDP effect algebras [7]), but OMLs lack RDP ( $MO_2$  witnesses), and the available image is only  $\sigma$ -epimorphic, not point-separating. The trivialising route is blocked, again non-distributively.

**Remark 4.2** (Segregated carriers do not witness). A carrier is *segregated* when its contextuality is carried by a central Boolean factor — e.g.  $MO_\omega$  or the product  $\prod_n MO_2$ , whose  $\sigma$ -additive two-valued states all concentrate at points. On such carriers every finitely coherent  $s_0$  is the restriction of a Dirac, so clause (ii) of Theorem 3.2 fails and there is no witness. A witness must be *non-segregated*: the incompatibility must be irreducible, not routed through a central partition. This is why Definition 1.1 requires irreducibility, and it sharpens the search for a carrier to the non-segregated, non-Boolean, non-Polish-representable regime.

## 5 The open problem

Assembling §3–§4, a  $\sigma$ -essential contextual state exists iff Question 3.6 has a positive answer on a carrier that is concrete,  $\sigma$ -complete, non-Boolean, irreducible, non-segregated, and non-Polish-representable. The boundary map of §4 settles the answer in no direction: none of the three natural bridges — the Blecher–Weaver dichotomy, a measurable cardinal, or a faithful representation — closes it.

**Question 5.1** (Main). Does there exist, in ZFC or under a large-cardinal hypothesis, a concrete  $\sigma$ -complete non-Boolean irreducible OML carrying a  $\sigma$ -essential contextual state? Equivalently, is the existence of a non-principal coherent  $\sigma$ -additive  $\{0, 1\}$ -selection across overlapping countable partitions (Question 3.6, non-segregated case) provable, refutable, or independent?

A resolution could take one of three forms, and the obstructions assembled above constrain each. A *positive* answer would exhibit a concrete carrier in the non-Polish, non-segregated regime; by Remarks 3.7 and 4.1 no admissible carrier assembles from countably many compatible pieces, so any witness must be uncountably generated and globally non-compositional. A *negative* answer would extend the Polish-representable emptiness of Derr–Williamson beyond its topological hypotheses to all concrete  $\sigma$ -OMLs. An *independence* result would supply two models, one with the selection and one without; the known precedents in the set theory of measure-theoretic structures—the von Neumann–Maharam problem [4] and real-valued measurable cardinals [8]—are all real-valued and strictly positive, leaving the two-valued, non-distributive case as the open edge.

To state the target precisely, write  $\text{St}(B)$  for the two-valued states on a sub-orthoposet  $B$ ,  $\text{St}_\sigma(\mathcal{L})$  for the global  $\sigma$ -additive two-valued states on  $\mathcal{L}$ ,  $\text{Fin}_\perp(\mathcal{L})$  for the finite  $\perp$ -closed sub-orthoposets, and  $\text{Adm}$  for the admissible carriers (concrete,  $\sigma$ -complete, non-Boolean, irreducible). A pattern  $s \in \text{St}(B)$  is realised by  $\mu$  when  $\mu \upharpoonright B = s$ . The  $\sigma$ -additive lifting property and its negation are

$$\begin{aligned}\Phi(\mathcal{L}) &:= \forall B \in \text{Fin}_\perp(\mathcal{L}) \forall s \in \text{St}(B) \exists \mu \in \text{St}_\sigma(\mathcal{L}) \mu \upharpoonright B = s, \\ \neg\Phi(\mathcal{L}) &:= \exists B \exists s \in \text{St}(B) \forall \mu \in \text{St}_\sigma(\mathcal{L}) \mu \upharpoonright B \neq s,\end{aligned}$$

the latter asserting a  $\sigma$ -essential contextual state (Def. 1.1, Thm. 3.2). The target sentence is

$$\Psi := \exists \mathcal{L} \in \text{Adm} \neg\Phi(\mathcal{L}) \equiv \exists \mathcal{L} \in \text{Adm} \exists B \exists s \forall \mu \in \text{St}_\sigma(\mathcal{L}) \mu \upharpoonright B \neq s.$$

Its leading quantifier ranges over carriers  $\mathcal{L}$  — families of subsets of  $\Omega$ , not natural numbers. By Proposition 2.1 every Boolean  $\mathcal{L}$  satisfies  $\Phi$ , so the witness is necessarily non-distributive; it is this second-order existential over an unbounded carrier, not the inner arithmetic, that places  $\Psi$  beyond any single combinatorial principle.

**Remark 5.2** (Constraints on a witness and on the available methods). The obstructions of §4 confine a witness to a sharply delimited regime and close the standard routes into it. Three boundaries combine. A Boolean carrier satisfies  $\Phi$  (Prop. 2.1); a Polish-representable carrier satisfies  $\Phi$  (Derr–Williamson); and a carrier whose contextuality is carried by a central Boolean factor is segregated, hence again satisfies  $\Phi$  (Rem. 4.2). A witness must therefore be simultaneously non-Boolean, non-Polish-representable, and irreducibly non-central: an uncountably generated, non-standard-Borel object whose obstruction is genuinely non-distributive. Two further boundaries bound the *methods*. Forcing cannot supply a witness in the existence direction: any constructible (in particular, Polish-representable) approximation falls under Derr–Williamson, so the object lies outside the reach of an explicitly described carrier, while set-forcing is consistency-strength preserving (Lévy–Solovay [15]) and so can frame independence but not manufacture the object. And the off-the-shelf representation machinery does not reach the carrier: the  $\sigma$ -Loomis–Sikorski theorem is established only under the Riesz decomposition property (RDP) [7], which a lattice effect algebra possesses if and only if it is an MV-algebra; an all-sharp non-Boolean OML lies exactly in the complement, where no  $\sigma$ -additive representation theorem is known to hold or to fail.

**Remark 5.3** (On the consistency strength of  $\Psi$ ). On a *Boolean* base the bare object of Question 3.6 is a non-principal countably-complete ultrafilter, hence a measurable cardinal; the non-distributive case is conjecturally at least as strong, and the analogue on  $B(\ell^2(\kappa))$  has exactly

Ulam-measurable strength [5]. It is tempting to read off that  $\Psi$  carries measurable-cardinal strength, but this is *not* established in either direction. The lower-bound argument—“no measurable cardinal, hence every  $\sigma$ -additive two-valued state is Dirac, hence no witness”—fails: its premise is an extension theorem for  $\sigma$ -algebras, not  $\sigma$ -classes, and is contradicted in ZFC by the Navara–Pták state (Example 3.4), which is non-Dirac; moreover, were the premise granted, it would render the non-Dirac clause vacuous and so point toward  $\Psi$  rather than against it. The upper bound is equally unearned: the relevant cardinal would govern two-valued  $(\{0, 1\})$  selections, where the real-valued-measurable machinery does not apply, and the Blecher–Weaver mechanism does not transfer (§4). The consistency strength of  $\Psi$  is therefore genuinely open—unsettled in both directions—rather than conjecturally measurable.

**Remark 5.4** (Localization of the open problem). Taken together, the boundaries locate the problem precisely while leaving no standard tool that enters it. A witness, if one exists, is an uncountably generated, non-Polish-representable, irreducibly non-central concrete  $\sigma$ -OML—and a survey of the natural candidate fields supplies no construction principle for such an object: non-separable and Maharam measure algebras are Boolean and real-valued; the descriptive set-theoretic no-section phenomena (non-smooth equivalence relations, co-analytic uniformization failure, turbulence) live over standard-Borel carriers with real-valued obstructions; the Dynkin-system and concrete-logic lineage either remains finitary or, in the Polish case, globalises. The closest structural rhyme is the Glimm–Effros dichotomy, under which non-smoothness of a countable Borel equivalence relation corresponds to the failure of a two-valued section; transporting that phenomenon onto a Gudder concrete logic would furnish exactly the missing principle, but no such transport appears in the literature, and constructing it would be original work. The honest status is thus a sharply located open cell: the witness question is neither closed by an impossibility theorem nor reachable by a borrowed construction, and its resolution awaits a genuinely new object inhabiting the non-Polish, non-distributive regime.

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